

Aditya Kulkarni

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Raleigh, NC (open to relocating to New York)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building and deploying Gen AI solutions end-to-end. At Isomer AI, built multi-model LLM workflows across Claude, GPT-4o, and Gemini with Temporal.io orchestration and MCP agent tooling, shipping 7 production deployments across 394 commits. Independently built AlgoPulse, a fault-tolerant agentic platform processing 2,000 async inference requests/second, and Glance AI, a compliance-ready RAG pipeline for fintech and healthcare. Fast at solving real-world problems under pressure: 3 complete AI MVPs in 48-hour builds, Top 10 across 100+ teams. Case studies: kulkarniaditya.com

CORE COMPETENCIES

Gen AI Workflow Deployment

LLM Systems in Production

End-to-End AI Delivery

ML Evaluation & Problem Decomposition

RAG Pipeline Design

Python & TypeScript

Distributed Systems

Client-Facing Technical Delivery

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Built end-to-end Gen AI workflows for a production claims automation platform: **Temporal.io workflow orchestration**, multi-model LLM routing across Claude, GPT-4o, and Gemini (3 providers, 8+ models) using MCP -- **394 commits, 7 production deployments**
- Delivered 10+ backend features including real-time event pipelines (Pusher), document processing (PDF generation, email parsing via Microsoft Graph API, S3), and role-based access control in TypeScript / Node.js / PostgreSQL / Redis
- Integrated Sentry observability and ran multi-model evaluations across 8+ variants to tune agent reliability and output quality in production

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed cloud-based genomic data platform (500GB, 100+ researchers worldwide) on Google Cloud Run; applied hyperparameter tuning and time-series ML analysis, improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built client-facing AI workflows for insurance customers including a health claim automation tool using OCR, NLP, and ML pipelines -- improved claim accuracy by 25%
- Supported model validation, red-teaming, and production reliability across AWS and GCP deployments with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a data processing pipeline automating support ticket analysis with Hadoop, Spark, and Kafka streaming -- reduced processing latency by 20%

PROJECTS

AlgoPulse -- Agentic AI Platform Independent, 2025

Identified API-polling bottleneck in a production AI orchestration system under load; re-architected to event-driven streaming with Supabase Edge Functions, Temporal.io durable workflows, and Kafka worker coordination. Delivered **2,000 async inference requests/second** with zero message loss. Real engineering problem, real fix, measured outcome.

Python, TypeScript, Node.js, Supabase Edge Functions, Temporal.io, Kafka

Glance AI -- Privacy-First RAG Pipeline Independent, 2025

Built a compliance-ready RAG system for fintech and healthcare clients with per-tenant data isolation at the vector retrieval layer. Prevents cross-tenant context leakage and LLM hallucinations -- production-ready for regulated enterprise environments.

Python, LLMs, Vector Databases, RAG

PerCent -- GenAI Finance App HackNC24 Top 10 / 100+ teams

Built a complete RAG-based GenAI finance app -- goal tracking, expense logging, Qwen-14B-Chat chatbot -- in 48 hours. Ranked Top 10 of 100+ teams. Demonstrates the same speed-to-production profile required for Palantir client deployments.

Python, Selenium, Hugging Face, Qwen-14B-Chat, RAG

Mapify -- ML and NLP Tool IEEE Published

T5-transformer pipeline converting 1,500+ PDFs into mind maps and flowcharts; reduced academic review time by 60%. IEEE published, presented to 500+ attendees. Demonstrates ML evaluation and problem decomposition from the ground up.

T5-transformer, NLTK, NetworkX, Python

EDUCATION

Master of Science, Computer Science North Carolina State University

Aug 2024 - May 2026

Courses: Artificial Intelligence, Machine Learning, Operating Systems, Data Science, Algorithms

B.Tech, Computer Science Mumbai University

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Machine Learning, Big Data Infrastructure, Blockchain, Algorithms

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C

AI/ML: LLM APIs (Claude, GPT-4o, Gemini), MCP, LangChain, RAG, Hugging Face, TensorFlow, PyTorch, ML Evaluation

Data & Infra: Temporal.io, Kafka, Spark, Hadoop, Distributed Systems, PostgreSQL, Redis

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD, Grafana, Sentry Frameworks: Node.js, React, Express, Prisma, Auth0